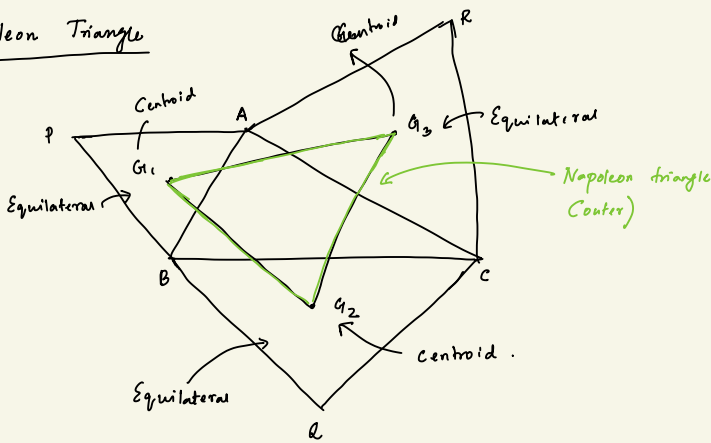
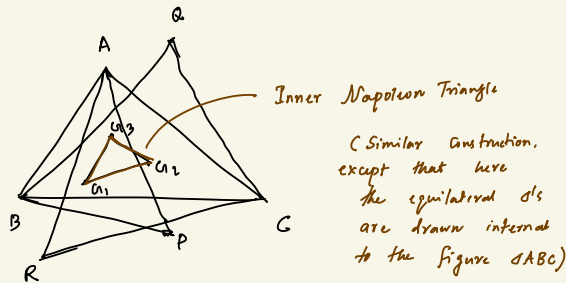


Napoleon Triangle



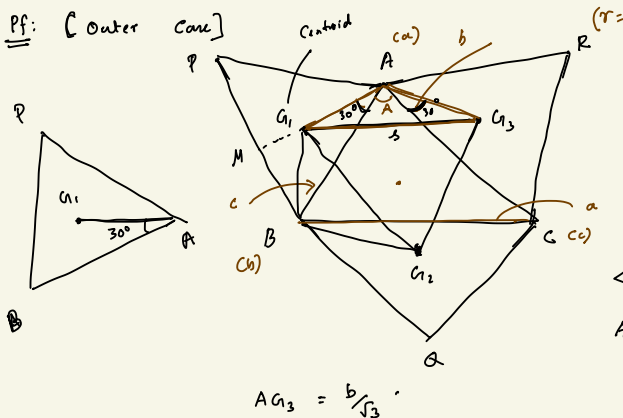
Let $\triangle ABC$ be any triangle. Equilateral triangles $\triangle PAB$, $\triangle QBC$, $\triangle RCA$ are drawn to the sides of $\triangle ABC$, external to it. Let G_1, G_2, G_3 be the centroid of $\triangle PAB, \triangle QBC, \triangle RCA$ respectively. Then $\triangle G_1G_2G_3$ is called the Napoleon Triangle (Outer) of $\triangle ABC$.

Inner Napoleon Triangle:



Result . The Napoleon triangle (both outer and inner) of any triangle is equilateral.

Pf: [Outer case]



$$G_1, G_3 = \frac{b}{\sqrt{3}}$$

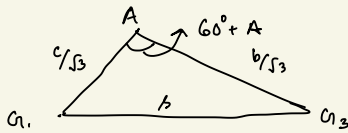
We consider $\triangle G_1AG_3$

$$\angle G_1AG_3 = 60^\circ + \angle A.$$

$$AG_1 = \frac{2}{3} AM.$$

$$= \frac{2}{3} \left(\frac{\sqrt{3}}{2} \cdot c \right)$$

$$= \frac{c}{\sqrt{3}}.$$



Now,

$$s^2 = \left(\frac{c}{\sqrt{3}}\right)^2 + \left(\frac{b}{\sqrt{3}}\right)^2 - 2 \cdot \frac{c \cdot b}{3} \cos(60^\circ + A)$$

$$= \frac{b^2}{3} + \frac{c^2}{3} - \frac{2bc}{3} [\cos 60^\circ \cos A - \sin 60^\circ \sin A]$$

$$= \frac{b^2}{3} + \frac{c^2}{3} - \frac{2bc}{3} \left[\frac{1}{2} \cdot \frac{b^2 + c^2 - a^2}{2bc} - \frac{\sqrt{3}}{2} \cdot \frac{2a}{bc} \right]$$

$$= \frac{b^2}{3} + \frac{c^2}{3} - \frac{2}{3} \left[\frac{b^2 + c^2 - a^2}{4} - a\sqrt{3} \right]$$

$$= \frac{2a\sqrt{3}}{3} + \frac{b^2}{3} + \frac{c^2}{3} - \frac{b^2 + c^2 - a^2}{6}$$

$$= \frac{2a}{\sqrt{3}} + \frac{2b^2 + 2c^2 - b^2 - c^2 + a^2}{6} = \left[\frac{2a}{\sqrt{3}} + \frac{1}{6} (a^2 + b^2 + c^2) \right]$$

$$\Rightarrow s = G_1 G_2 = \sqrt{\frac{2a}{\sqrt{3}} + \frac{1}{6} (a^2 + b^2 + c^2)}$$

$$= G_2 G_3 = G_1 G_3$$

So, $\Delta G_1 G_2 G_3$ has to be equilateral. $\square$

$\square$ For inner Napoleon triangle, each side length = ω

$$\omega^2 = \frac{2a}{\sqrt{3}} - \frac{1}{6} (a^2 + b^2 + c^2) \quad \checkmark$$

Result: $\ar(\text{Outer Napoleon triangle}) + \ar(\text{Inner Napoleon triangle})$
 $= \ar(\Delta ABC)$

Pf: $\ar(\text{Outer Napoleon triangle}) = \frac{\sqrt{3}}{4} s^2$

$$= \frac{\sqrt{3}}{4} \left[\frac{2a}{\sqrt{3}} + \frac{1}{6} (a^2 + b^2 + c^2) \right]$$

$$= \frac{a}{2} + \frac{\sqrt{3}}{24} (a^2 + b^2 + c^2)$$

Similarly, $\ar(\text{Inner Napoleon triangle}) = \frac{\sqrt{3}}{4} \omega^2$

$$= \frac{\sqrt{3}}{4} \left[\frac{2a}{\sqrt{3}} - \frac{1}{6} (a^2 + b^2 + c^2) \right]$$

$$= \frac{a}{2} - \frac{\sqrt{3}}{24} (a^2 + b^2 + c^2)$$

So, $ar(\text{Outer NT}) + ar(\text{Inner NT})$

$$= \frac{a}{2} + \frac{\sqrt{3}}{24} (a^2 + b^2 + c^2) + \frac{a}{2} - \frac{\sqrt{3}}{24} (a^2 + b^2 + c^2)$$

$$= a : ar(\triangle ABC) \quad \square.$$

□ Weitzenbock's Inequality

In any $\triangle$, with side length a, b, c , and area $= \Delta$,

$$\Delta \geq \frac{\sqrt{3}}{12} (a^2 + b^2 + c^2).$$

Ans: In any $\triangle$, let the side length of the inner NT be w .

Then,

$$\left\{ \frac{2\Delta}{\sqrt{3}} - \frac{1}{6} (a^2 + b^2 + c^2) \right\} = w^2 \geq 0$$

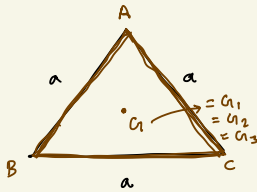
$$\Rightarrow \frac{2\Delta}{\sqrt{3}} \geq \frac{1}{6} (a^2 + b^2 + c^2)$$

$$\Rightarrow \Delta \geq \frac{\sqrt{3}}{12} (a^2 + b^2 + c^2) \quad \square.$$

$$\Rightarrow \frac{\Delta}{a^2 + b^2 + c^2} \geq \frac{\sqrt{3}}{12}$$

$\text{Min} \left(\frac{\Delta}{a^2 + b^2 + c^2} \right) = \frac{\sqrt{3}}{12} \rightarrow \text{Equality if } a=b=c.$

□



In case when the main triangle $\triangle ABC$ is equilateral, the inner NT becomes just a Pt $\rightarrow G$ (Centroid of $\triangle ABC$)
 $\Rightarrow$ Side length of inner NT = 0

□ The centroid of $\triangle ABC$ (actual triangle) and of $\triangle G_1 G_2 G_3$ (Napoleon) is the same.